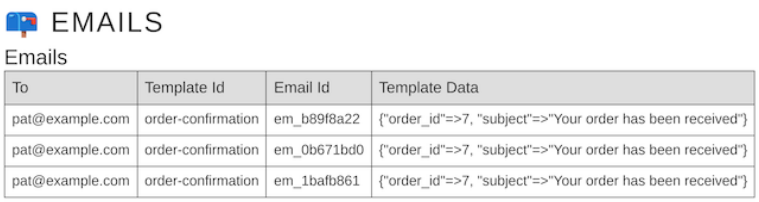
Addressing Idempotence Issues with Third Parties

The fake email service behaves in the way described above: when you ask it to crash, it's configured to send an email first and *then* crash. If you navigate to its UI at `http://localhost:3001/email/ui`, you'll see that there are multiple emails to the same person for the same order:



A screenshot of the fake email UI showing multiple emails sent to the same person for the same order.

This is harder to recover from: the part of our system that is told about email delivery is the part that's failing. You can't know if the email was sent—only the email service knows that. There are two common ways to manage this type of failure. One is to ask the service if the operation completed.

Use Third Party Services APIs to Figure Out What Happened

Most third-party services allow you to query the service to find out what operations have been previously performed. The fake email service provides a way to do this, and you can use that to figure out if an email you are about to send has already been sent.

In the case of the example app, only one email confirmation should be sent per order. Thus, the combination of email address, `template_id` and order ID should be unique. If you query the email service and it returns a sent email with those attributes, you can be sure the email you want to send has been sent already, and skip it. If no result is returned, trigger the service to send it. See the diagram that follows.